

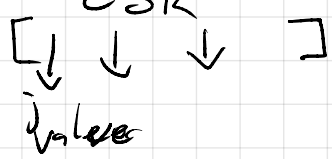
COO

i

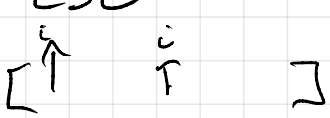
j

values

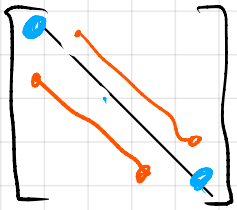
CSR



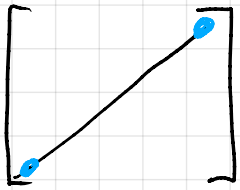
CSC



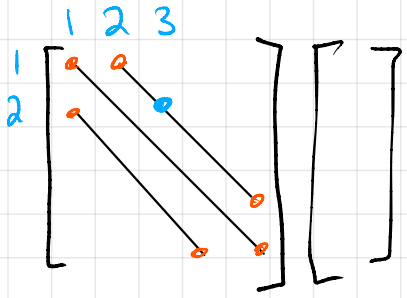
Structure exploitation



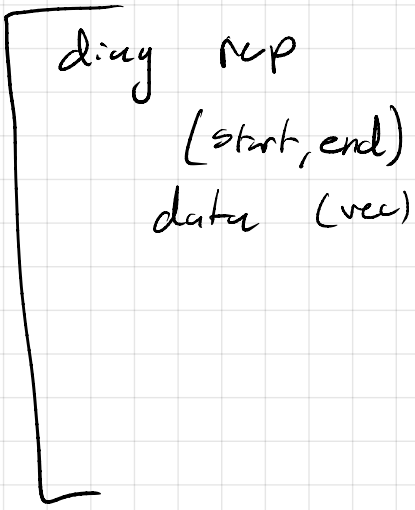
$i = j$



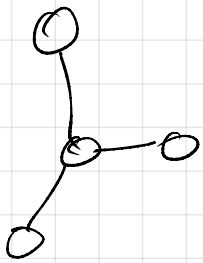
$i = N - j$

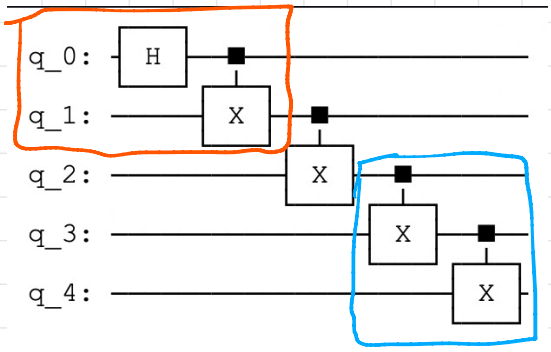


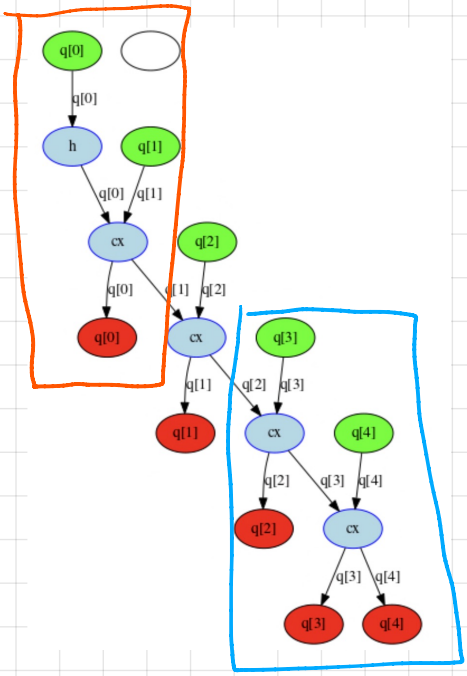
[2, 3]



$H \otimes H$







$$(CNOT)(H \otimes I)(q_0 \otimes q_1)$$

GOAL: Memory Efficiency
(not contraction choices)

$$[(CNOT)[(H \otimes I)(H \otimes I)(H \otimes I)(H \otimes I)](q_0 \otimes q_1)]$$

New struct:

start i, j
end k, l

data : [, , , , , , ,]

complex "integers"

a + bi w/ a, b ∈ ℤ

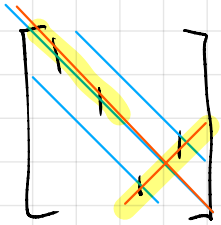
a, b ∈ {-1, 0, 1}

Most existing pkgs use numpy

Dense ⇒ ~~struct?~~ ↳ Class?

$$\begin{bmatrix} 1 & 2 & 0 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \Rightarrow$$

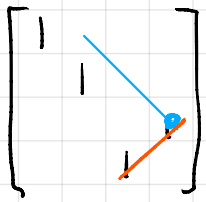
Assume principal diagonals

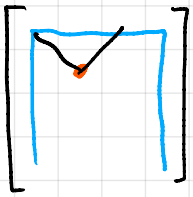


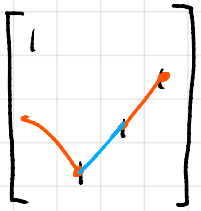
$H H^*$ do we need constants?

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

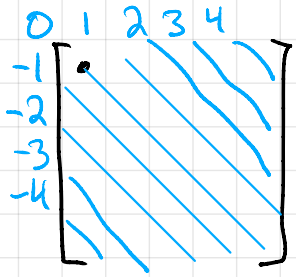
$\sqrt{1+1} = \sqrt{2}$

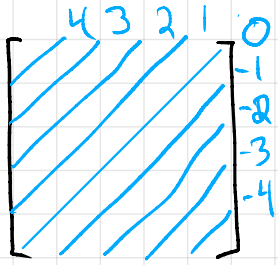






10/28





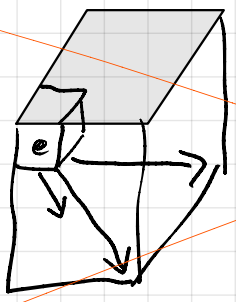
$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$$

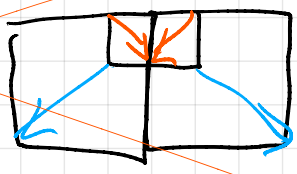
only use principal diags

(sparsity ratio criteria?)

$$\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$

consider diagonal sparsity? later





Stick to 2D! (don't need more for QC)

TODOs:
- Matrix (complexes) (srikar)
- Vector class (Blake) (and cleanup general struct)
- start kernels (M·V (Blake), M·M (srikar), M⊗M later...)
- finish Dense/sparse conversions (srikar)

M·V $\begin{bmatrix} 1 & \\ & \end{bmatrix}\begin{bmatrix} 2 \\ 1 \end{bmatrix}$

M·M $\begin{bmatrix} & \\ & \end{bmatrix}\begin{bmatrix} & \\ & \end{bmatrix}$

Maybe?

$a \cdot b = a^T b$

$\langle a | b \rangle$

Shoot to have these done by 11/1

$A \otimes B \begin{bmatrix} A_{11}[B] & A_{12}[B] \\ A_{21}[B] & A_{22}[B] \end{bmatrix}$